

**Rochester Institute of Technology  
Department of Computer Engineering  
CMPE 630 Digital IC Design**

**Final Project:  
ALU with BIST**

**Individual Final Project**

**Due: 05/02/2025**

CMPE 630 Digital IC Design  
Final Project

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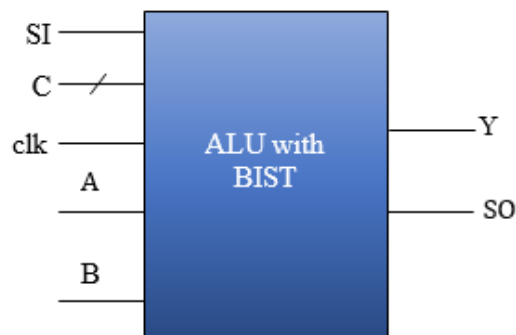
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## 1. Introduction

This project will help solidify students' understanding of basic digital IC design techniques through the design, implementation, and verification of an ALU. The ALU will compute the following:

$$Y = \begin{cases} A + B, & sel = 0 \\ A * B, & sel = 1 \end{cases}$$

The two operand inputs, **A** and **B**, are 8-bits. The interface of your design should look like Figure 1, with the signals described in Table 1.



**Figure 1. ADD/MULT ALU with BIST.**

**Table 1. ALU Interface Signals.**

| Signal | Description                                  |
|--------|----------------------------------------------|
| A,B    | 8-bit Input vector                           |
| clk    | Clock                                        |
| SI     | Scan-in                                      |
| SO     | Scan-out                                     |
| Y      | Output vector                                |
| C      | Control signals for enabling test mode, etc. |

You will need to pick a specific VLSI optimization for your design. If there is something you are interested in (or you need to do for your project/thesis), pitch it! I am open to most any optimization you are interested in. I will also allow some alterations to the project itself to accommodate thesis/project research goals. **I must approve your selected optimization by 4/11/2025!**

**Some Possible Ideas (no need to pick from this list):**

|                         |                           |                                                 |
|-------------------------|---------------------------|-------------------------------------------------|
| Dynamic Voltage Scaling | Dynamic Frequency Scaling | An IEEE-Published Adder/Multiplier Architecture |
| RAZOR                   | Multiple V <sub>th</sub>  | Multiple V <sub>dd</sub>                        |
| Voltage Islands         | Power Gating              | Inter-Module Pipelining                         |
| Watermarking            | SRAM Scratchpad           | Logic Locking                                   |

## 2. Milestone 1 – Project Proposal

1. Do some research on the architecture of BIST and ALU. Plan and sketch out a block diagram of your design. Think about what major components (e.g. Adder, Multiplier, Signature Generator...) you will use, what cells are needed, and how to make those components.
  - a. Note that there might be some standard cells you may need which are missing from your library.
2. Think about what optimization you are using and how to implement it.
3. Prepare a 3 to 4 minute presentation going over your design. The recommended presentation structure is listed below.
  - a. Cover slide, which includes your name, and course number
  - b. Overall block diagram of your design (one slide)
  - c. Major components of your design (one slide per major component)
  - d. Your optimization (one slide)
4. Presentation will be held in class on 04/14/2025
5. Milestone 1 due 04/11/2025

### Milestone 1 Submission:

Submit your slides and a short 3-5 sentence description of your optimization to MyCourses.

## 3. Milestone 2 – ALU

1. In Lab 8, you should have completed a schematic and layout of an 8-bit full adder using gates from your standard cell library. The result is an adder circuit with 16 input pins and 9 output pins. If this has not yet been completed, place the cells and auto-route following steps similar to Lab 8. Note the following:
  - a. VDD and VSS pins have to be on M1
  - b. You **must** honor the defined horizontal/vertical wiring direction per metal layer
  - c. Your layout should consist of more than one row. You will also need to adjust the PR calculation spreadsheet based on the cell dimensions
  - d. You **should verify the functionality** of your adder after you have designed the schematic and before the layout. This can be done by driving a set of stimuli into your adder and verifying its output.
  - e. Ensure your adder passes DRC, LVS, and QRC
2. Using gates from your standard cell library create the schematic of an 8-bit multiplier in a new cell at one hierarchical level (library cell FML\_MULT). Place the cells and auto-route following steps similar to Lab 8. The result is a multiplier circuit with 16 input pins and 16 output pins.
  - a. VDD and VSS pins have to be on M1
  - b. You **must** honor the defined horizontal/vertical wiring direction per metal layer
  - c. Your layout should consist of more than one row. You will also need to adjust the PR calculation spreadsheet based on the cell dimensions

- d. You **should verify the functionality** of your multiplier after you have designed the schematic and before the layout. This can be done by driving a set of stimuli into your multiplier and verifying its output
  - e. Ensure your multiplier passes DRC, LVS, and QRC
3. Using gates from your standard cell library create the schematic of **an 8-bit ALU** in a new cell at one hierarchical level (library cell **FML\_ALU**). Place the cells, and auto-route following steps similar to Lab 8. The result is an ALU circuit with **17 input pins and 16 output pins**
  - a. VDD and VSS pins have to be on M1
  - b. You **must** honor the defined horizontal/vertical wiring direction per metal layer
  - c. Your layout should consist of more than one row. You will also need to adjust the PR calculation spreadsheet based on the cell dimensions
  - d. You **should verify the functionality** of your ALU after you have designed the schematic and before the layout. This can be done by driving a set of stimuli into your multiplier and verifying its output.
  - e. Ensure your ALU passes DRC, LVS, and QRC
  - f. See **Figure 2** for an example ALU layout.
4. Test the pre-extracted and extracted ALU circuit for worst case delay and maximum clock frequency.
5. **Milestone 2 due 04/23/2025**

### Milestone 2 Submission:

Submit a **3-part report** with each part discussing and including the following per module (Adder, Multiplier, and ALU). This does not need to be a standard report. **This does not need to be a full report.** Submit relevant **screenshots/codes/calculations with captions** under the corresponding headers.

- Logic Function (e.g.  $Y = A + B$ )
- Schematic capture
- Layout capture
- Cell dimension (width, height, and area)
- Functional simulation waveform (**must show at least 8 passing cases**)
- Passing DRC, LVS, and QRC screenshots
- Extracted layout
- **\*\*For ALU only:** worst-case path delay and maximum clock frequency as well as **how you calculate it**
- Relevant codes (e.g. Verilog/VHDL test benches if you use mix-signal simulation)

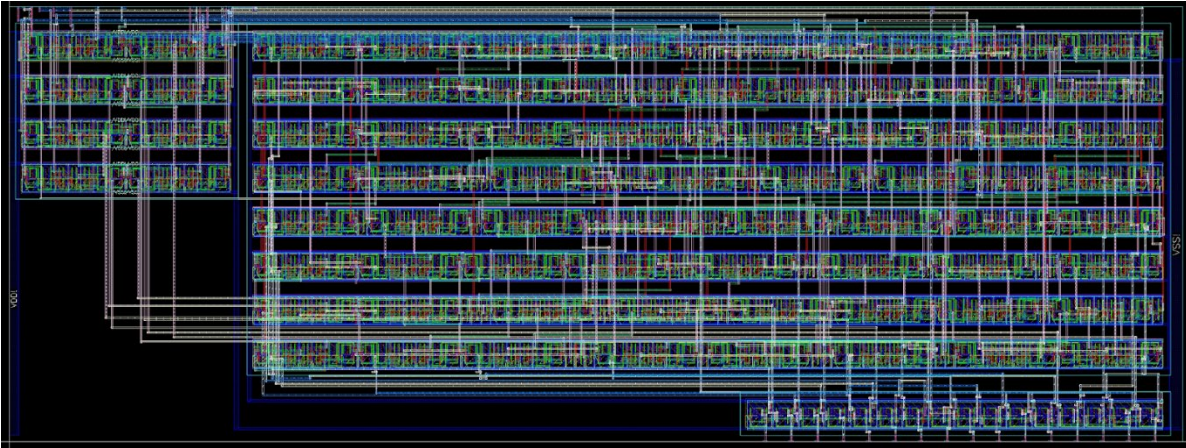


Figure 2. Example ALU Layout

#### 4. Milestone 3 – Built In Self-Test (BIST)

1. Following similar procedures to milestone 2, make the missing cells needed for the BIST. The final design (ALU with BIST) should be able to perform the following tasks if **test mode is enabled**:
  - a. Capture an input seed from the test controller.
  - b. Generate and feed pseudo random stimuli to your DUT.
  - c. Capture and feed the DUT output to signature generator.
  - d. Generate and report the signature to the test controller.
  - e. *Note that the above is a generic description for a BIST, the functional testing performed for the project is stated in point 6 below.*
2. You will create your **test controller as a Verilog testbench**. You will need to capture **two** ALU with BIST schematics: **one without** the test controller and **one with** the test controller. Follow instructions similar to Lab 8 for creating a Verilog Testbench and applying it to a design. You will perform a **mixed-signal simulation using the test controller (i.e., your testbench) and ALU with BIST to perform the functional testing** (see point 6).
3. The ALU with BIST functions as a normal ALU if normal operation mode is enabled (i.e. **test mode is NOT enabled**).
4. You may use more than one bit for your control signal to handle different modes of operation and test (e.g., loading seed, capture DUT output, feed signature, check signature, etc.).
5. **You are responsible for generating the golden signature** according to your design. A python file has been provided with the project to help you generate this signature. Your golden signature also needs to account for the optimization you chose if it impacts the design's behavioral functionality.
6. **Functional Testing**: Verify the functionality of your design with BIST using **seed A = 10110011 and B = 00111011**, ensuring the reported signature matches the golden signature after 32 clock cycles.
7. Tested the pre-extracted and extracted circuit **in normal operation mode** for worst-case delay, maximum clock frequency.
8. **Milestone 3 – Due 05/02/2025**

### Milestone 3 Submission:

Submit a **report** including the following information for your BIST. **This does not need to be a full report.** Submit relevant **screenshots/block diagrams/calculations with captions** under the corresponding headers.

- Functional description/block diagram
- Golden signature used and how it is calculated
- Schematic capture
- Layout capture
- Cell dimension (width, height, and area)
- Functional simulation waveform (**must show a BIST pass**)
- Passing DRC, LVS, and QRC screenshots
- Extracted layout
- Relevant codes (e.g. Verilog/VHDL test benches if you use mix-signal simulation)

## 5. Cells routing guidelines:

1. VDD and VSS pins must be on Metal 1 for all your blocks
2. **You are not routing VDD and VSS.** After generating the initial placement as taught in Lab 8, manually place the VDD and VSS pins and associated labels on top of a standard cell VDD/VSS bus.
  - a. For cells with multiple rows, manually insert the power strapping before routing. To efficiently constraint the router, it may be necessary to move the power strapping outside the PR Boundary Box; as necessary, you will need to account for the power strapping in your area calculation.
3. To restrict the VDD and VSS nets from being auto routed, open the Navigator (**Window->Assistants->Navigator**), select all nets except VDD and VSS, and then next to **Route Net** press **Selected** to only route the nets you selected.
4. Once routing is finished run DRC and LVS. If there is a small number of DRC violations, go ahead and fix them by hand, however, only fix by hand as a last resort and when you are completely finished with routing as running the router again will remove your changes.
5. Suggested routing channel height:
  - a. 2 tracks top and bottom
  - b. 4 tracks between rows
6. Since you will be using these cells with other routed cells. Optimize the pin placement (top, bottom, left, and right) and minimize routing when you place cells together as you build larger blocks.
7. Left/Right side VDD/VSS strapping should be at least  $30\lambda$  wide
8. If DRC returns hundreds of “soft check” errors, note that “soft check” errors are ESD violation. This indicates the design is missing well/substrate ties or you’ve confused the tools by missing a VDD or VSS pin, label, or connection somewhere

## 6. Final Project Report

Your project report should be at least 5 pages IEEE Transactions style with at least the following sections. **This should be a full and comprehensive report.**

1. Abstract
  - Describe what was done, how you did it, and the significant results
2. Introduction
  - What is the project about?
  - What is the background?
3. Preliminaries
  - What is your selected design optimization?
  - How does it work and what is the goal?
  - Relevant background + resources used to implement it.
4. Design Methodology
  - Discuss your design approach. In particular:
    - i. Did you have a fully parallel or serial design, or some hybrid?
    - ii. Was your design driven by optimal area or optimal performance, or both?
  - Include block diagrams of your design.
  - Cover these same topics for your design optimization
5. Results/Data Analysis
  - Present and analyze the functional and timing results.
6. Conclusions
  - Restate important results, how they were obtained, and any uncertainties, possible improvements, or implications
7. Appendices
  - Waveforms and other diagrams that are too large for the body of the paper.

## 7. Final Project Grading

Table 2: Final Project Grading

|                            |      |
|----------------------------|------|
| Milestone 1 Score          | 10%  |
| Milestone 2 Score          | 40%  |
| Milestone 3 Score          | 30%  |
| Final Project Report Score | 20%  |
| Total Score                | 100% |